

Lab 3.1 – Location Tools

At the end of this lab exercise, Participants will be able to:

1. Locate the word COGNEX on the demo plate

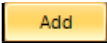
The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

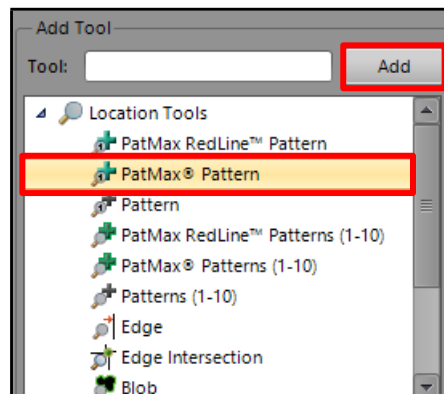
2. Easy Builder User Interface Application Step 2
 - Locate Part

Follow the steps below to complete the lab exercise:

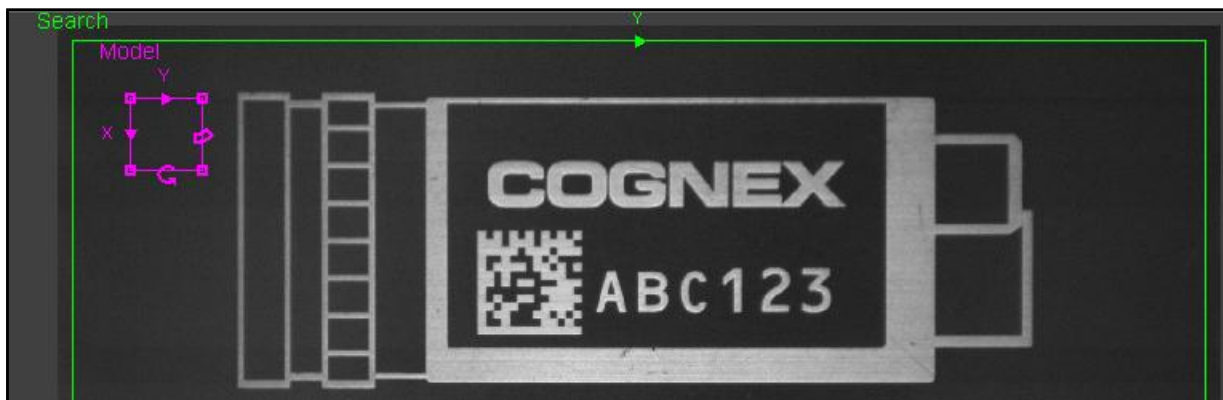
1. Load the **MyScaleCalibration** job.

NOTE: Confirm the demo plate is on the PASS side in the Image view window.

2. Save the job as **MyLocator** in the folder on your desktop created in Lab #2.
3. Click the **Locate Part**  Application Step button.
4. Select **PatMax® Pattern** in the Location Tools and click the **Add**  button.

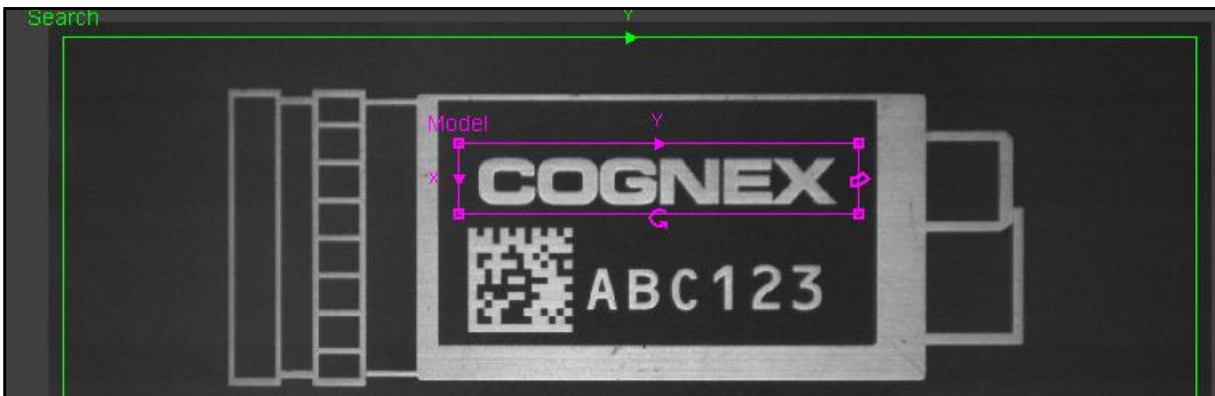


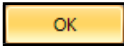
A **Search and Model** box will appear in the Image window.

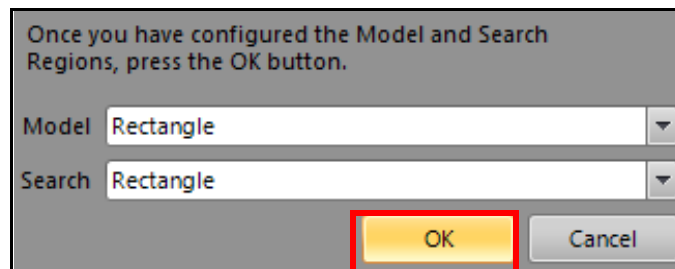


5. Click on the model Region and move over the **COGNEX** logo.

NOTE: You do not need to move the Search region or you can click the Maximize Region tool  to maximize the search region.



6. Once you have configured the region, click the **OK**  button.



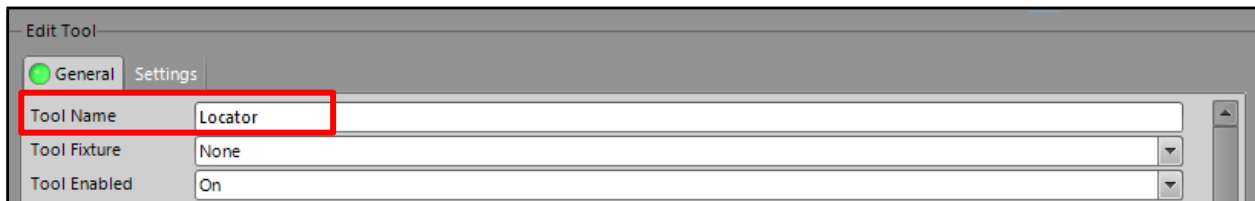
7. Confirm in the **Image view** that the search box and the pattern that was taught is outlined in green and a green cross is in the center of the trained image on the live part.



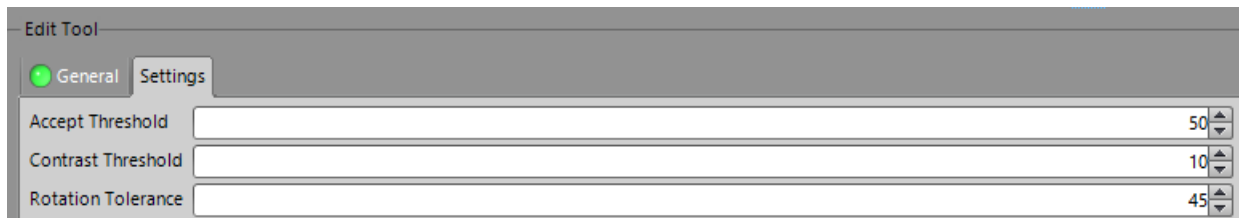
8. Confirm the COGNEX logo is visible in the **Trained Image** tab.



9. Change the Tool Name from Pattern_1 to **Locator**.



10. Click the Settings tab and change the **Rotation Tolerance** from 15 to 45.



11. Click the Manual Trigger (or <F5>) and rotate the part in the Image view window.
12. Confirm that the Locator tool crosshair repositions itself to the center of the COGNEX logo with each trigger.

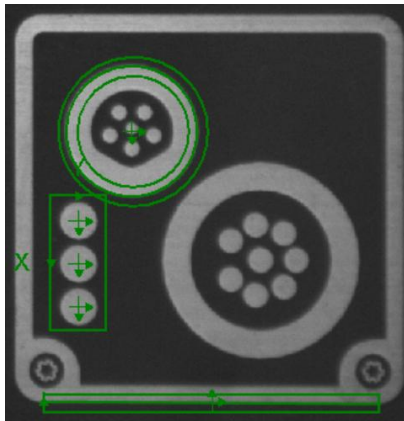


NOTE: The Learned pattern must be *WITHIN* the search area for the tool to Pass as seen in the screen shot above.

13. Change the Rotation Tolerance to **90** and move your part in the FOV – review the results.
14. Change the Rotation Tolerance to **180** and move your part in the FOV – review the results.
15. Save the job.

Lab 3.2 – Exercises

1. Continue with previous job – **MyLocator**, can you use **Blobs (1-10)**, **Circle**, and **Edge** tools to locate the three blobs, the outer circle and the bottom edge as shown below? After adding all the tools, verify if they work for any rotation angles for both **Pass** and **Fail** demo plates. Take a screenshot of the **Image View** and the **Palette** (e.g., by pressing **Shift + Windows key + S**), similar to the screenshot shown on the last page. Save the screenshot and submit it on **Blackboard**.



2. In Lab 3.1, you used PatMax Pattern tool to locate the part. Can you try using PatMax RedLine Pattern instead? Take a screenshot of the **Image View** and the **Palette**. Save the screenshot and submit it on **Blackboard**.
3. In Lab 3.1, you used a paper that contains one single demo plate. Now can you put both plates under the camera, including the **Pass** and the **Fail** plates, and start a new job to locate both “COGNEX” logos? Take a screenshot of the **Image View** and the **Palette**. Save the screenshot and submit it on **Blackboard**.
4. You will use emulation for this question. **Save and submit your job on Blackboard.**
 - 1) Use In-Sight **5400 Color** emulator. Start a new job.
 - 2) Download the Faces file from Blackboard and extract it.
 - 3) Browse the images in the Faces folder.
 - 4) Use Image13.bmp to train the model.
 - 5) Adjust parameters and verify if the tool finds the person in every image.
 - 6) **Save the job as MyFaces_YourName.job and submit it.**

Screenshot Example:

